In [1]:
```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier # Or KNeighborsClassifier
from sklearn.metrics import accuracy_score

df = pd.read_csv("titanic_train.csv")

features = ["Pclass", "Sex", "Age", "Fare"]
target = "Survived"

X = df[features].copy()
y = df[target]

X["Sex"] = X["Sex"].map({"male": 0, "female": 1})

median_age = X["Age"].median()
X["Age"] = X["Age"].fillna(median_age)

X["Fare"] = X["Fare"].fillna(X["Fare"].median())

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_sta

print(f"Training set size: {len(X_train)}")
print(f"Test set size: {len(X_test)}")
```

```
Training set size: 623
Test set size: 268
```

In [2]:
```
clf = DecisionTreeClassifier(max_depth=3, random_state=42)
clf.fit(X_train, y_train)

y_pred = clf.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)

print(f"Test Set Accuracy: {accuracy:.4f}")
```

```
Test Set Accuracy: 0.8060
```

Model Error Analysis: The most common mistake made by the model regards middle-aged male passengers with a mid-level fare because their survival was more influenced by luck than by clear trends like class or gender.